

Cybersecurity Incident Report

Section 1: Identify the type of attack that may have caused this network interruption

One potential explanation for the website's connection timeout error message is: A TCP SYN flood attack is the most likely explanation for the website's connection timeout error message.

The logs show that:

The logs show that many TCP SYN requests were repeatedly sent from the IP address 203.0.113.0 to the company's web server. As the volume of SYN requests increased, legitimate users experienced connection failures, reset packets, and HTTP 504 Gateway Timeout errors.

This event could be:

This event could be classified as a Denial of Service (DoS) attack because the malicious traffic originated from a single source IP address. The attacker attempted to overwhelm the server by flooding it with TCP SYN packets, preventing legitimate users from accessing the website.

Section 2: Explain how the attack is causing the website to malfunction

When website visitors try to establish a connection with the web server, a three-way handshake occurs using the TCP protocol. Explain the three steps of the handshake:

- Step 1: The client sends a TCP SYN packet to the server to request a connection.
- Step 2: The server responds with a SYN-ACK packet to acknowledge the request and indicate that it is ready to establish a connection.
- Step 3: The client sends an ACK packet back to the server, completing the TCP three-way handshake and establishing the connection.

During a TCP SYN flood attack, the attacker sends a large number of SYN packets but does not complete the final ACK step. As a result, the server keeps numerous half-open connections active and allocates resources to each incomplete connection. Over time, these half-open connections consume server resources and prevent legitimate users from accessing the website, resulting in connection timeout errors and degraded network performance.

Explain what happens when a malicious actor sends a large number of SYN packets all at once:

When a malicious actor sends many SYN packets all at once, the server must allocate resources to respond to each request. As the number of requests increases, the server's available resources become exhausted.

Explain what the logs indicate and how that affects the server:

The logs indicate repeated SYN packets from a single external IP address targeting port 443 (HTTPS) on the web server. The unusually high volume of connection requests suggests an attempt to overwhelm the server and prevent legitimate users from establishing connections. As the attack continued, legitimate users experienced failed connections, reset packets (RST, ACK), and HTTP 504 Gateway Timeout errors. Eventually, the server became unable to respond to normal user requests because its resources were consumed by the attack traffic.

This attack negatively affected network performance by slowing or preventing access to the company's website. Employees and customers could not reliably access vacation package information, resulting in service disruption and reduced business availability. Potential mitigations include implementing SYN flood protection, configuring firewall rate limiting, deploying intrusion detection and prevention systems (IDS/IPS), enabling SYN cookies, and using traffic filtering or DDoS protection services to identify and block malicious traffic.